Lecture 06

30.10.2025

We will often use the short hand notation

$$L = -\sum_{i,k=1}^{n} a_{ik}(\cdot)\frac{\partial^2}{\partial x_i \partial x_k} + \sum_{i=1}^{n} b_i(\cdot)\frac{\partial}{\partial x_i} + c(\cdot)$$

and write $Lu = f$.

<u>Remark</u> the classification of linear second-order PDEs is similar to the ones for classification of quadratic equations in 2D:

$$\frac{x^2}{a} + \frac{y^2}{b} = c \quad \rightsquigarrow \text{ describes an ellipse for } a, b, c > 0$$

$$-\left(\frac{1}{a}u_{xx} + \frac{1}{b}u_{yy}\right) = f \quad \rightsquigarrow \text{ describes an elliptic PDE for } a, b > 0$$

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$$\frac{x^2}{a} - y = 0 \quad \rightsquigarrow \text{ describes a parabola}$$

$$-\left(\frac{1}{a}u_{xx} - u_y\right) = 0 \quad \rightsquigarrow \text{ describes a parabolic PDE}$$

$$\frac{x^2}{a} - \frac{y^2}{b} = c \quad \rightsquigarrow \quad \text{describes a hyperbola for } a, b, c > 0$$

$$-\left(\frac{1}{a} u_{xx} - \frac{1}{b} u_{yy}\right) = c \quad \rightsquigarrow \quad \text{describes a hyperbolic PDE for } a, b, c > 0$$

<u>Remark</u> We also called the linear transport equation $u_t - \underset{\neq 0}{a} u_x = 0$ "hyperbolic" - why?

By taking the partial derivative w.r.t. x we obtain

$$\frac{\partial}{\partial t}(u_t - a u_x) = \frac{\partial}{\partial t} 0$$

$$\Rightarrow \quad u_{tt} - a u_{xt} = 0 \qquad (*)$$

By taking the partial derivative w.r.t. t instead, we have

$$\frac{\partial}{\partial x}(u_t - a u_x) = \frac{\partial}{\partial x} 0$$

$$\Rightarrow \quad u_{tx} - a u_{xx} = 0 \qquad (**)$$

Combining $(*)$ and $(**)$ then yields

$$u_{tt} \overset{(*)}{=} a u_{xt} = a u_{tx} \overset{(**)}{=} a(a u_{xx}) = a^2 u_{xx}$$

$$\Rightarrow \quad u_{tt} - a^2 u_{xx} = 0 \quad \Rightarrow \text{diffusion matrix } \begin{bmatrix} -1 & 0 \\ 0 & \underbrace{a^2}_{>0} \end{bmatrix}$$

$$\Rightarrow \text{hyperbolic PDE!}$$

In particular, this means that solutions of $u_t - a u_x = 0$ also solve the (wave) equation $u_{tt} - a^2 u_{xx} = 0$.

Analogus considerations can be done for $u_t + a u_x = 0$.

This reflects that the wave operator $\frac{\partial^2}{\partial t^2} - a^2 \frac{\partial^2}{\partial x^2}$ splits into

$$\frac{\partial^2}{\partial t^2} - a^2 \frac{\partial^2}{\partial x^2} = \left(\frac{\partial}{\partial t} - a \frac{\partial}{\partial x}\right)\left(\frac{\partial}{\partial t} + a \frac{\partial}{\partial x}\right).$$

Well-posedness of PDEs

Motivation: for our numerical schemes we want to compute approximations $\tilde{u}$ to "the" exact solution

$\Rightarrow$ this only makes sense if there <u>is</u> a solution and we can only reasonably compare $\tilde{u}$ and u if there are not multiple solutions ($\rightsquigarrow$ uniqueness!)

Finally, if the PDE / problem is very sensitive w.r.t. perturbations (e.g. roundoff error), numerical schemes will most likely fail

We therefore look for PDEs that are <u>well-posed</u> in the sense of <u>Hadamard</u>:

a) Existence of a solution

b) <u>Uniqueness of solutions</u>

c) Stability: the solution behavior changes continuously with the given data

Regarding <u>uniqueness of solutions</u> consider the three prototypical PDEs

1) $$-\Delta u(x) = f(x)$$

2) $$u_t(t,x) - \Delta u(t,x) = f(t,x)$$

3) $$u_{tt}(t,x) - \Delta u(t,x) = f(t,x)$$

As stated, none of these equations has a unique solution. Assume for example that $u(t,x)$ is <u>a</u> solution to 2). We can then define <u>arbitrarily many other solutions</u>

by defining

$$v(t,x) := u(t,x) + c$$

where $c \in \mathbb{R}$ is a constant value. Indeed, it then follows

$$v_t(t,x) - \Delta v(t,x)$$

$$= \frac{\partial}{\partial t}\left(u(t,x) + c\right) - \Delta\left(u(t,x) + c\right)$$

$$= u_t(t,x) + \underbrace{\frac{\partial}{\partial t} c}_{=0} - \Delta u(t,x) - \underbrace{\Delta c}_{=0}$$

$$= u_t(t,x) - \Delta u(t,x) \underset{\substack{\uparrow \\ \text{since } u \text{ is} \\ \text{a solution!}}}{=} f(t,x)$$

$\rightsquigarrow$ we generally need <u>more</u> equations for uniqueness

<u>Definition</u> (boundary conditions) Let $\Omega \subseteq \mathbb{R}^n$ be a domain $\Gamma \subset \partial\Omega$ a part of the boundary.

i) We call the condition

$$u = g \quad \text{on } \Gamma$$

<u>Dirichlet boundary condition</u> (for $g = 0$ homogeneous). If the first coordinate has the interpretation of time and $\Gamma = \{0\} \times \Omega$, then we call the condition an <u>initial condition</u> instead

ii) We call the condition

$$\nu \cdot \nabla u = g \quad \text{on } \Gamma$$

<u>Neumann boundary condition</u> (for $g = 0$ homogen.).

Here $\nu : \Gamma \to \mathbb{R}^n$ denotes the outer normal vector field.

iii) We call the condition

$$\alpha u + \beta \nu \cdot \nabla u = g \quad \text{on } \Gamma$$

<u>Robin / mixed boundary condition</u> (for $g = 0$ homogeneous).

<u>Remark</u> there exist other kinds of boundary conditions (periodic / far field boundary conditions) which we do not specify here in more detail.